University of Glasgow
School of Computing Science

Complete Figures and Results

Blockchain-Based Electronic Voting: Bribery, Sybil and Ballot-Stuffing Attacks

Every figure and every result table produced by the simulation, collected in one document. Part I presents the 10 figures at full size with their captions and a note on what each one shows. Part II reproduces the 14 result tables in full. Part III records the headline numbers and the reproducibility protocol.

Generated from the simulation output at seed 42 over 2,500 paired trials. Every number is injected from `results/` at build time; none is typed by hand. Companion to the main dissertation.

Contents

Part I. Figures **2**

Part II. Result Tables **12**

Table 1. Parameter ranges used in the randomised trials 12

Table 2. Bribery attack — success probability (single gate: detection is folded into $1 - q$) 12

Table 3. Sybil attack — both measures: surviving (headline) and accepted-only 13

Table 4. Ballot stuffing — both measures: surviving (headline) and accepted-only 13

Table 5. Baseline versus protected — headline measure only (agrees with Tables 7–9) 13

Table 6. Chernoff bound validation statistics 13

Table 7. Statistical analysis of the paired design 14

Table 8. Countermeasure effectiveness (absolute and relative reduction) 15

Table 9. Overall attack comparison across the three vectors 15

Table 10. Sample scenarios with the manipulation threshold m 16

Table 11. Critical thresholds T1–T7 per benchmark scenario 17

Table 12. Optimal attacker effort per vector (G1) 18

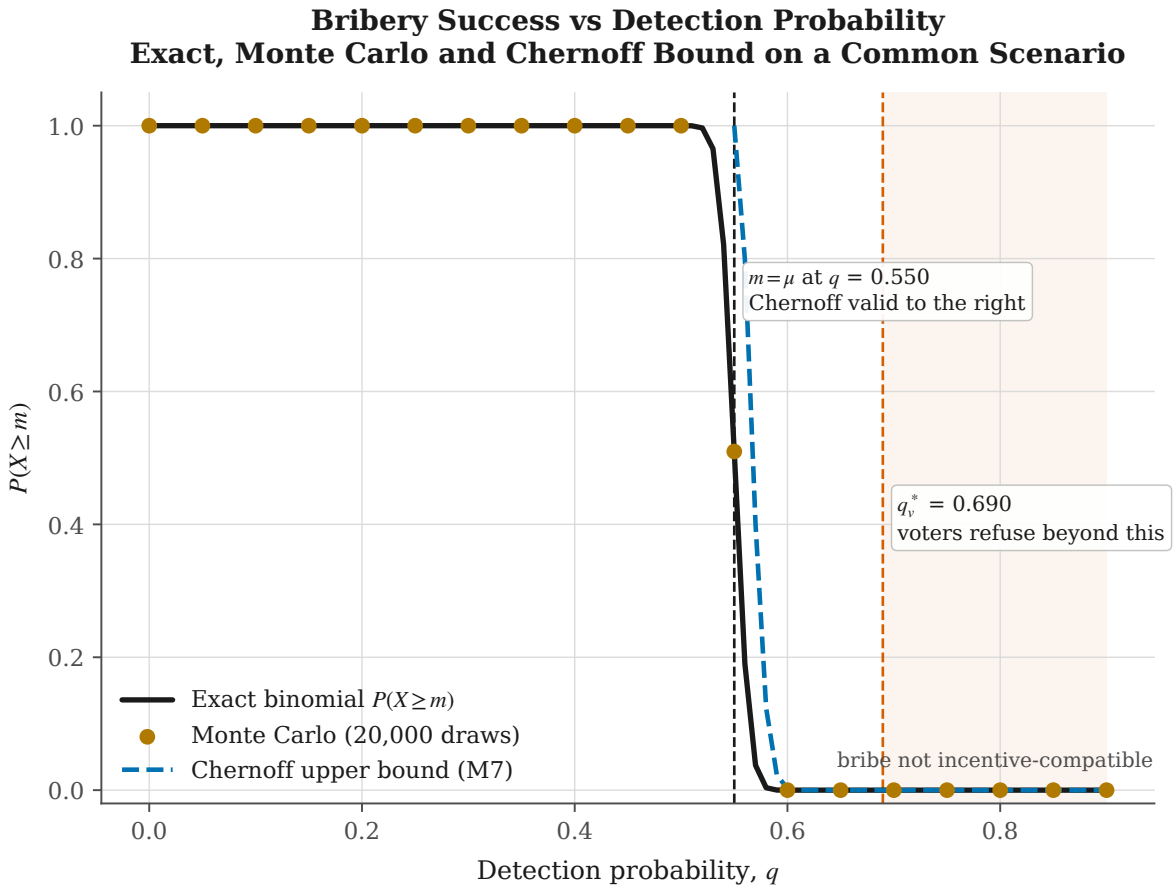
Table 13. Coalition attack versus best single vector (G2) 19

Table 14. Sensitivity of the coalition finding to cost assumptions 20

Part III. Headline Numbers and Reproducibility **23**

Part I. Figures

Each figure below is the one that appears in the dissertation, at full size, with its caption unchanged. The shaded note beneath each states what the figure is for and which threshold values are marked on it.



Fixed scenario: N = 2,000, m = 900, B = 4,000,000, R = 2,000, P = 900, giving T = 2,000 attempts. Only q varies.

Figure 1. Exact binomial, Monte Carlo and Chernoff bound against detection probability on one fixed scenario. Averaging these three quantities across heterogeneous randomised trials would compare them on different sub-populations — the bound exists only where $m > \mu$ — so this figure varies only q.

How to read it. Three independent methods on one scenario. The exact binomial curve and the Monte Carlo markers coincide throughout; the Chernoff curve sits above both wherever it is defined, which is the validity check. Where the bound disappears, $m \leq \mu$ and the one-sided upper tail does not apply. R and P are chosen so the collapse sits to the LEFT of q_v^*, i.e. inside the region where a rational voter would still accept the bribe.

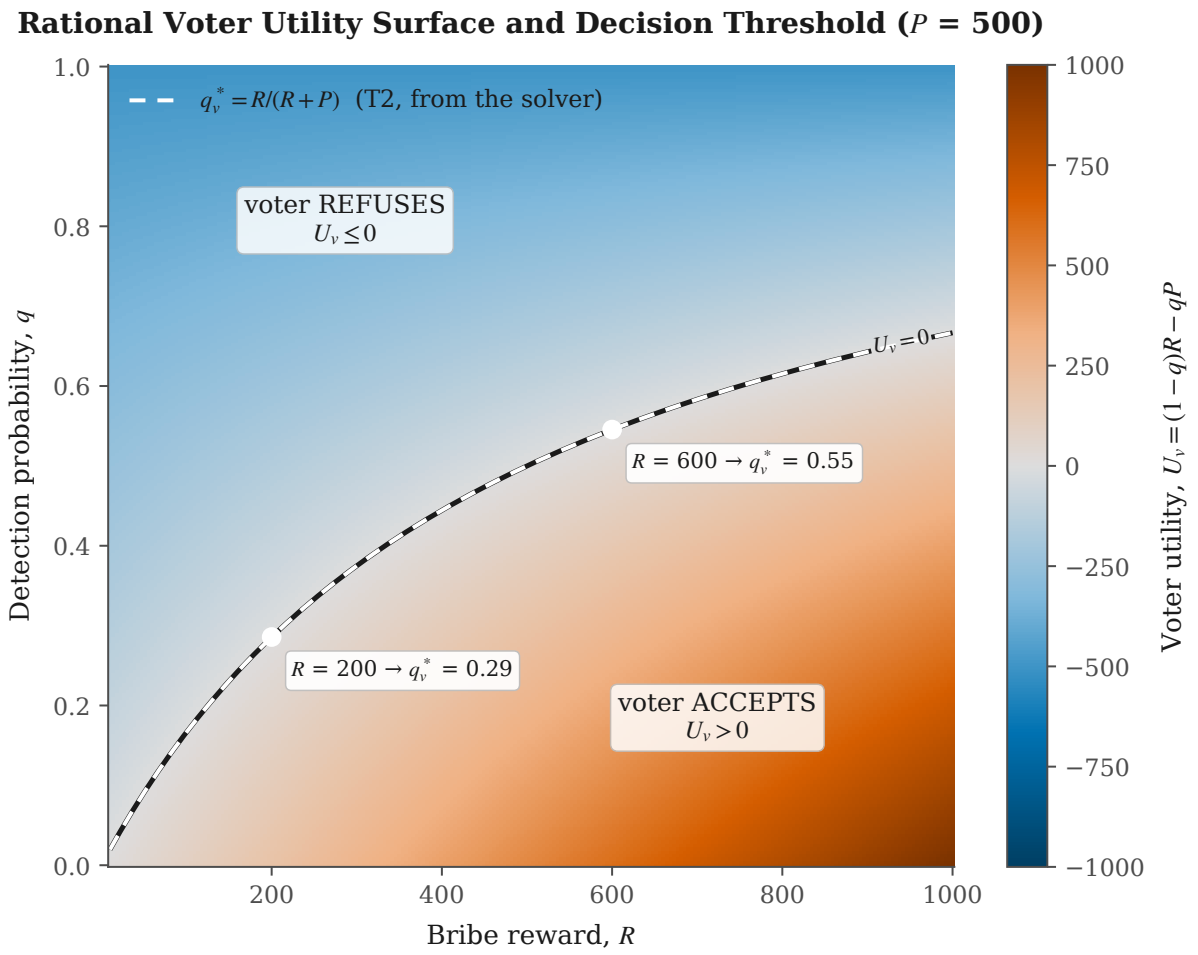
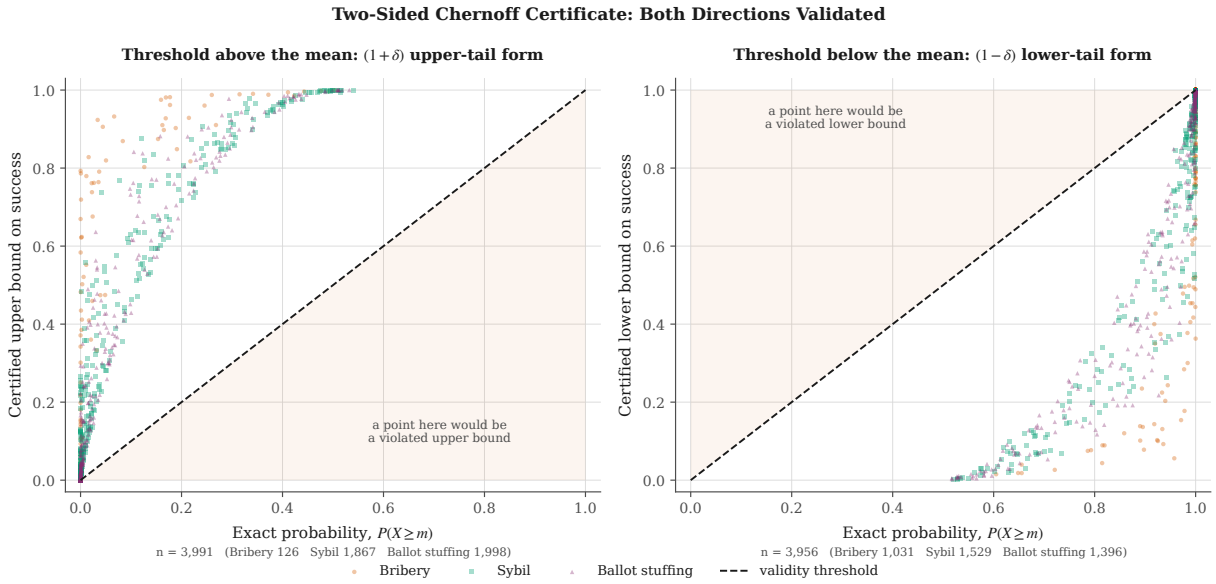


Figure 2. Rational voter utility $U_v = (1-q)R - qP$, with the $U_v = 0$ decision boundary and the closed-form indifference threshold $q_v^* = R/(R+P)$ drawn together.

How to read it. The rational-voter gate. A voter accepts a bribe only while $U_v = (1-q)R - qP$ stays positive; the crossing at $q_v^* = R/(R+P)$ is the point beyond which no budget buys a vote, because the voter — not the system — refuses.



Every one of the 15,000 scenario evaluations receives a certified statement, and the certified interval contains the exact value in 15,000 of 15,000 cases. The panels plot the 7,947 evaluations where a Chernoff bound does the work; the remaining 7,053 are exact by construction (the threshold is at or beyond the support, so the probability is 0 or 1 and no bound is needed) and would sit on the diagonal's endpoints. The one-sided bound alone covers only the left panel.

Figure 3. Both Chernoff directions validated. Left: threshold above the mean, where the $(1+\delta)$ upper-tail form certifies success from above — valid points lie on or above the diagonal. Right: threshold below the mean, where the $(1-\delta)$ lower-tail form applied to the failure event certifies success from below — valid points lie on or below it. No point falls on the wrong side of either diagonal.

How to read it. The two-sided Chernoff certificate. Every scenario now carries an analytic guarantee: an upper bound where $m > \mu$, and a lower bound obtained by applying the lower tail to the failure event where $m \leq \mu$. The exact value lies inside the certified interval in 100% of evaluations.

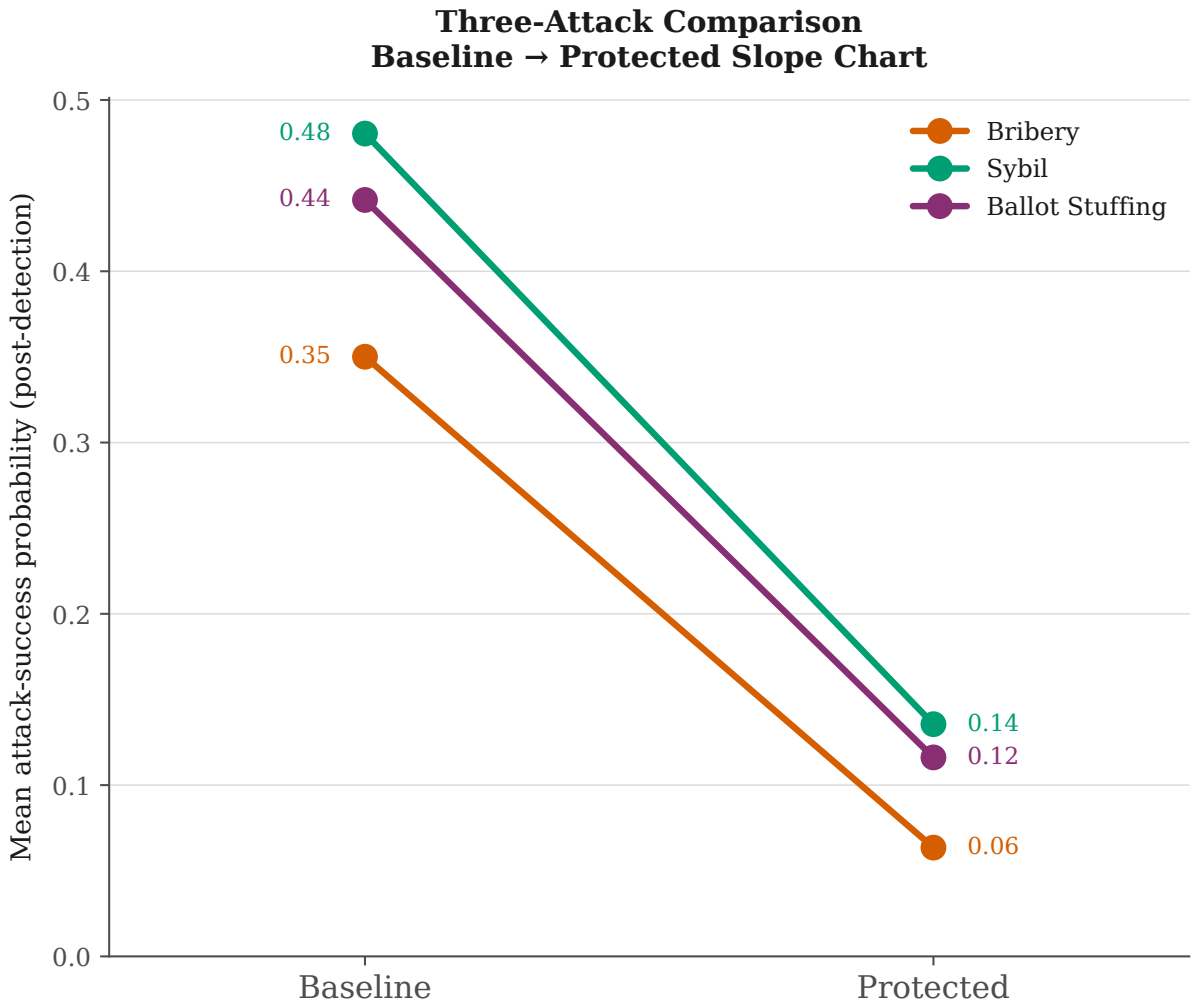
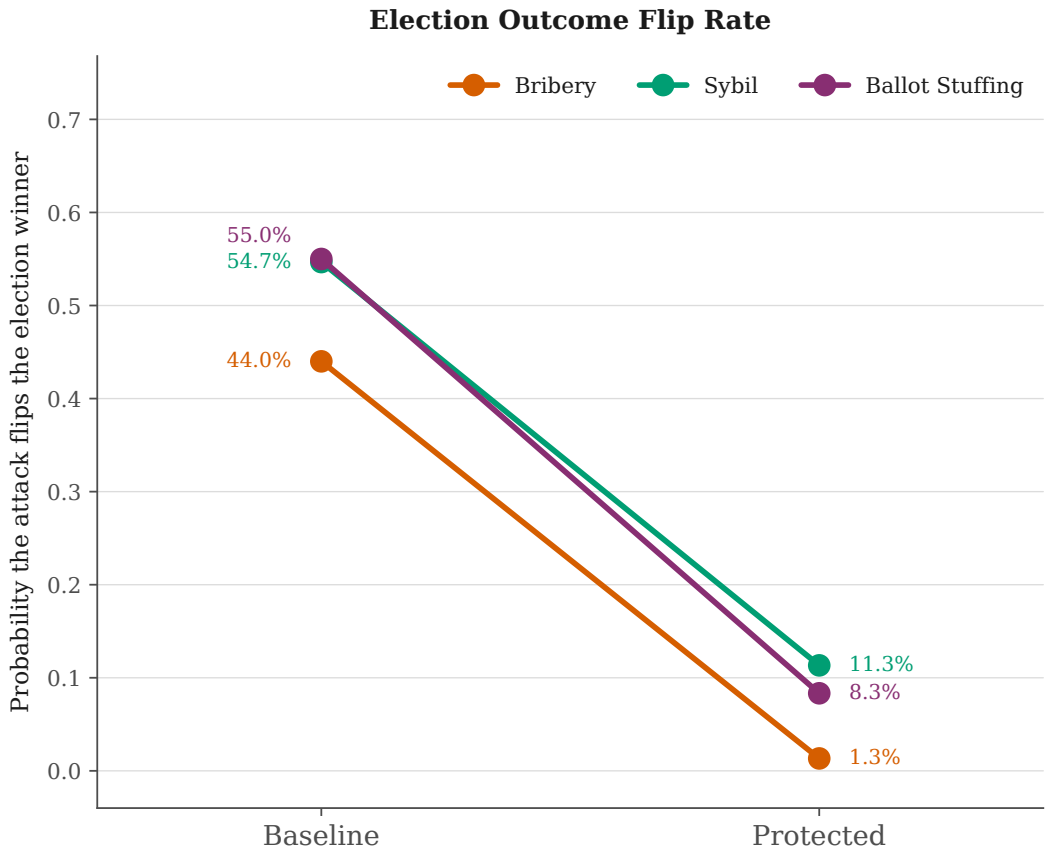


Figure 4. Mean success probability for the three vectors, baseline against protected.

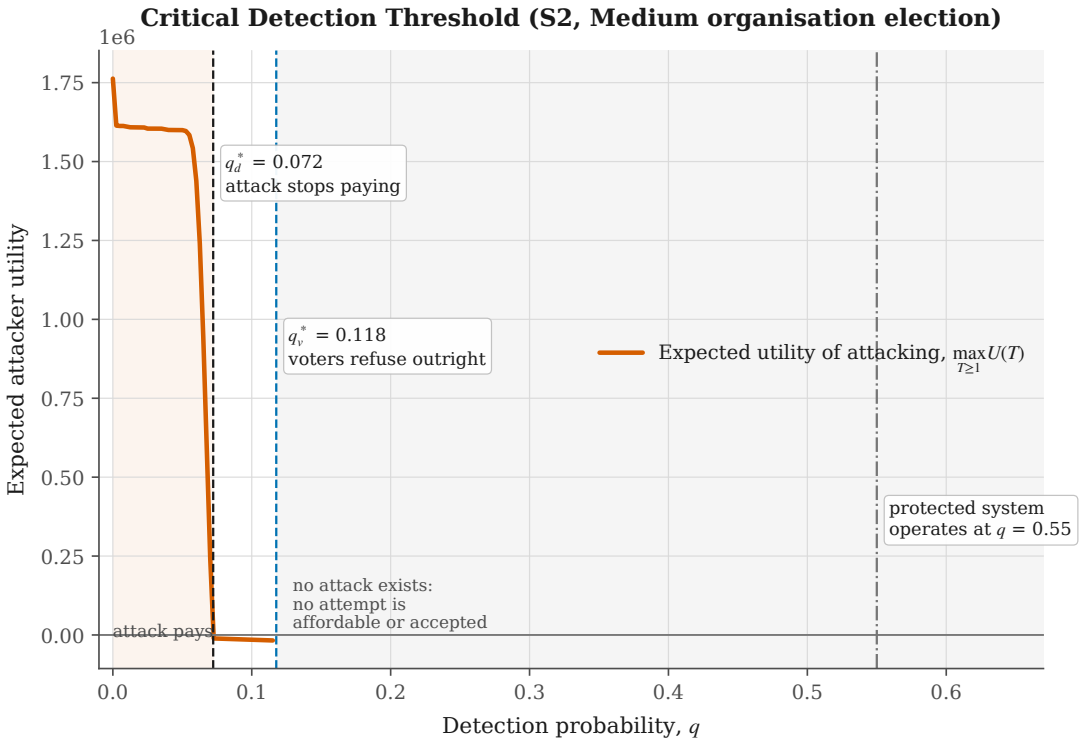
How to read it. The headline comparison. Baseline against protected, per vector, on identical paired trials. The gap between the two bars is the countermeasure effect; the error bars are bootstrap intervals, used because the underlying distributions are bimodal rather than normal. Note that the protected bars are reduced, not zero: attack success is substantially lowered under the modelled countermeasures, not eliminated.



Measured on genuinely mined chains with real tallies and real winner determination, not on abstract success probabilities.

Figure 5. Probability that an attack flips the winner of a concrete, blockchain-recorded election.

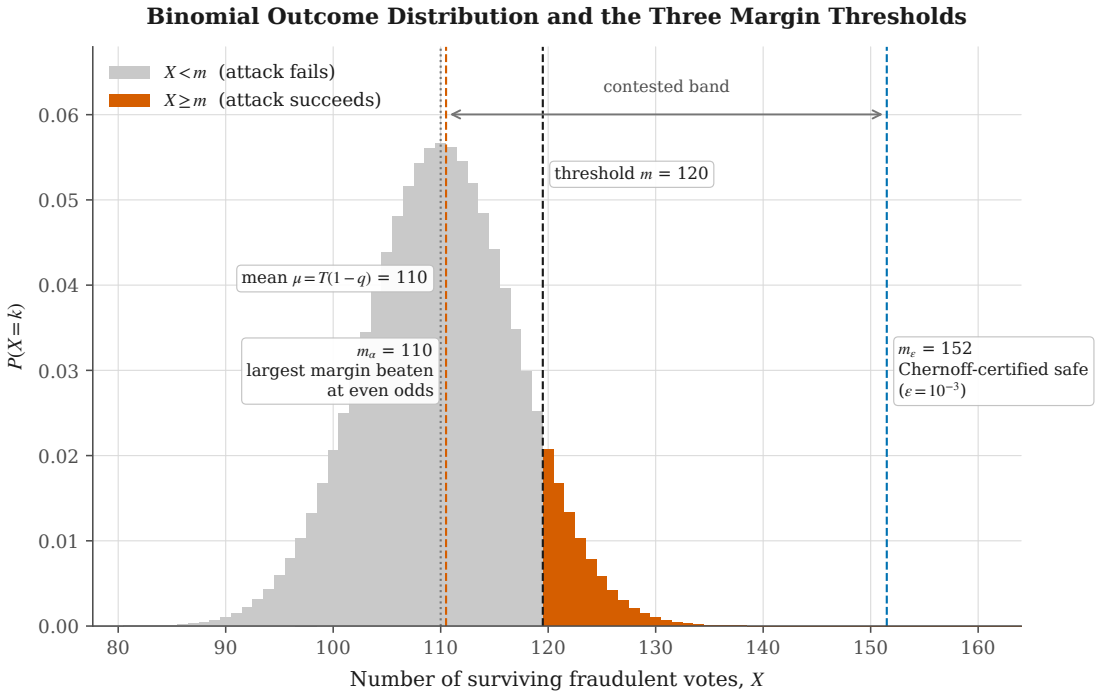
How to read it. The concrete layer. Abstract success probability is one thing; the rate at which a mined election actually changes winner is another. This figure shows the two agree in magnitude, in the size of the protection effect, and in placing bribery lowest. They do not agree on the ordering of the top two: Sybil leads in the analytic layer and ballot stuffing here, by a third of a percentage point — inside sampling error, so those two are best read as indistinguishable rather than ranked.



Only about 7.2% detection is needed to make this attack unprofitable, well below the 11.8% at which voters would refuse outright, and far below the 55% the protected configuration actually achieves.

Figure 6. Expected utility of attacking as detection rises, for the medium-organisation scenario. Two thresholds are marked: q_d^*, where the attack stops paying, and q_v^*, where voters refuse outright.

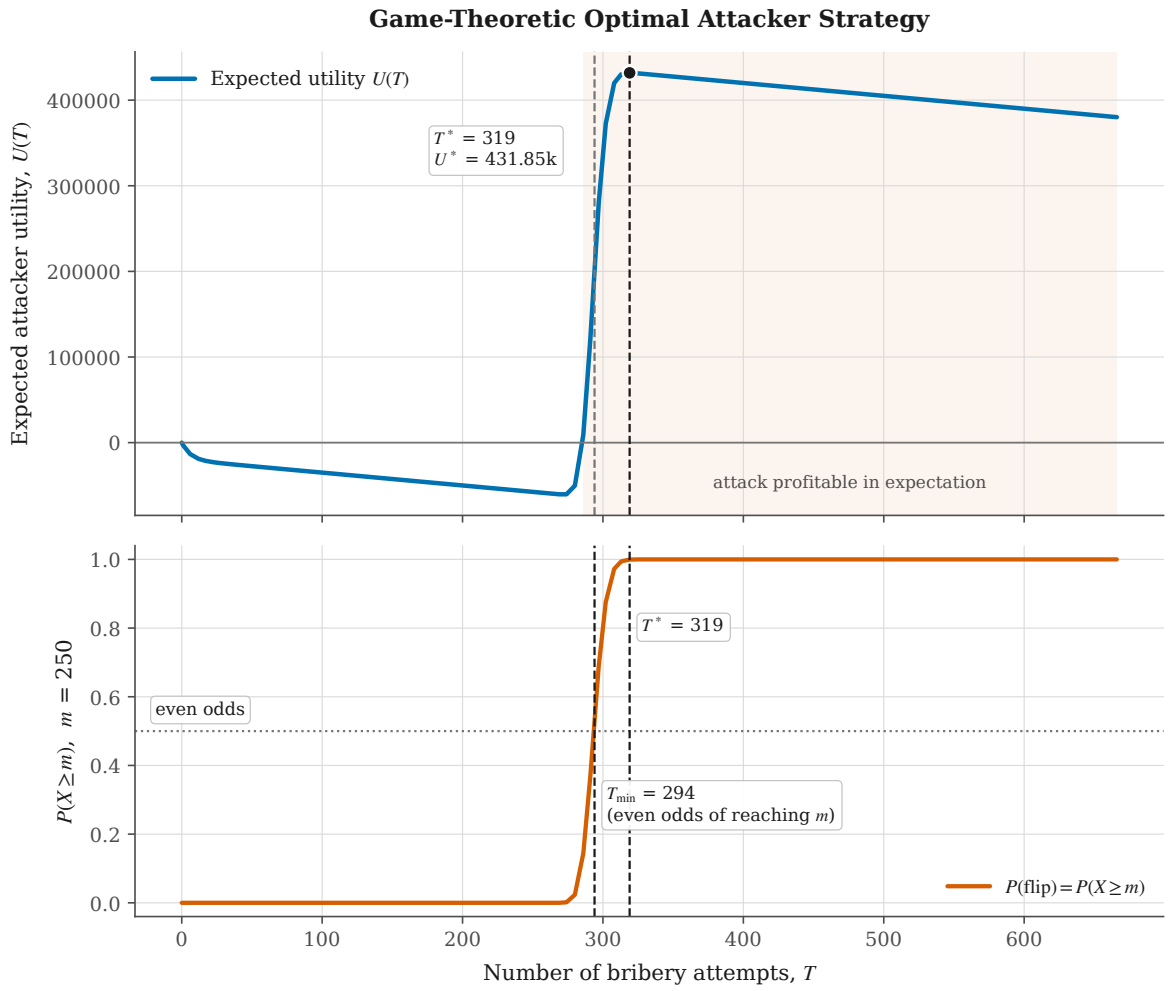
How to read it. The defender's design target. q_d^* is where attacking stops paying — the attacker still can, but no longer wants to. q_v^* is where voters refuse outright. The gap between them is the region where deterrence relies on the attacker being rational. The protected configuration's much higher operating point is a deliberately generous margin in the model, not a level the analysis finds necessary — far less detection would already deter.



$X \sim \text{Binomial}(T = 200, 1 - q = 0.55)$; the shaded success region has probability $P(X \geq m) = 0.088$. Because X is integer-valued, region boundaries are drawn at half-integers (the m line sits at $m - 0.5$), so each line meets the edge of a bar rather than cutting through its centre. Margins at or below m_α are comfortably within reach; margins at or above m_ε are provably out of reach.

Figure 7. The binomial outcome distribution with its mean, the manipulation threshold, the attacker-reachable margin and the Chernoff-certified safe margin all marked.

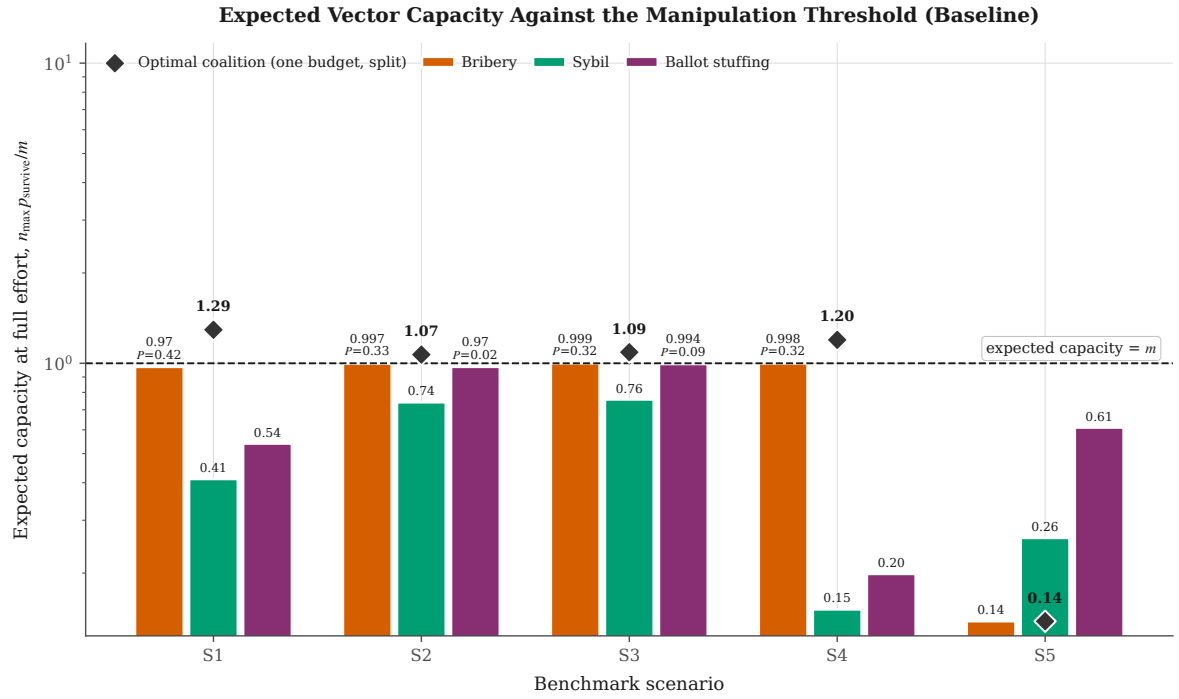
> **How to read it.** The distribution behind every other figure, with all three margin thresholds drawn on it. Region boundaries sit at half-integers because X is integer-valued, so each line meets the edge of a bar rather than cutting through its centre.



Fixed scenario: $N = 2{,}000$, $m = 250$, $B = 100{,}000$, $R = 150$, $q = 0.15$. Utility peaks at $T^* = 319$ and then declines: once the flip probability saturates, each further bribe adds cost and detection risk but no additional chance of success. T^* is the best point on the evaluated grid rather than an exact argmax; the curve is flat near its peak, so nearby values of T are near-optimal too.

Figure 8. Expected attacker utility (top) and flip probability (bottom) against bribery attempts, with the utility-maximising T^* and the even-odds effort threshold $T_{\min}$ marked.

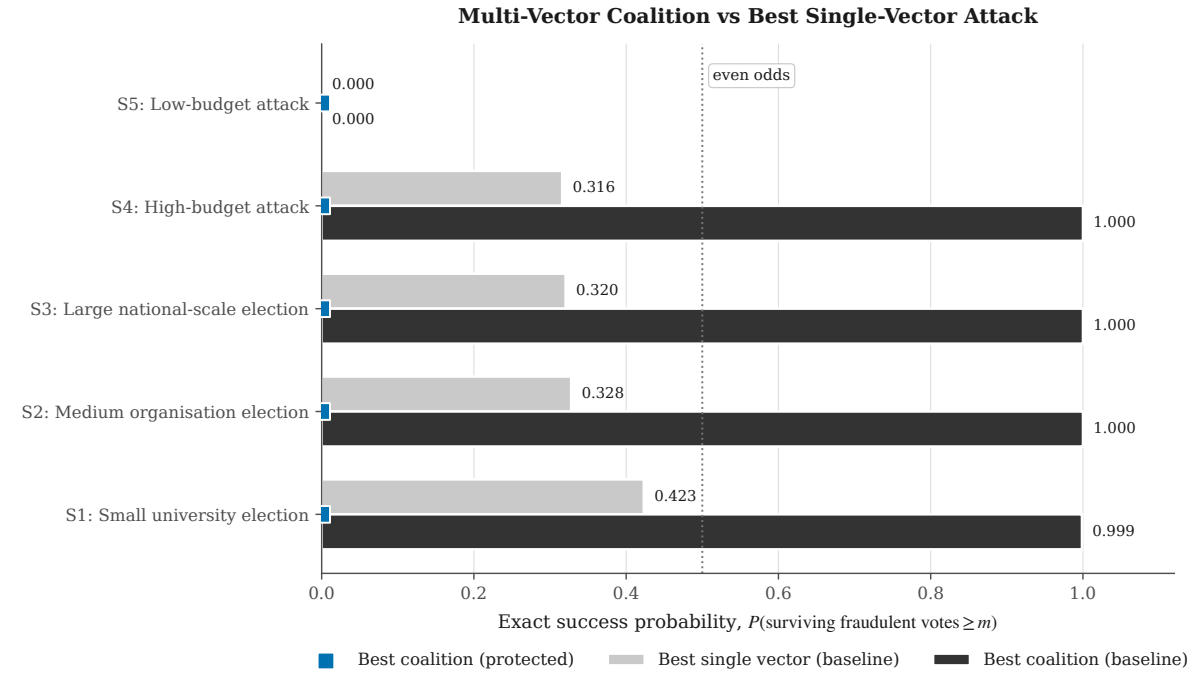
> **How to read it.** The attacker's optimisation. Utility against number of attempts, with the optimum marked. Effort beyond the optimum lowers utility, which is why a budget-constrained attacker does not simply spend everything.



Bars are EXPECTED surviving votes over the threshold, so a bar below the line means the mean falls short -- not that the threshold is out of reach. *P* is the exact success probability at full effort, and it shows how quickly that distinction stops mattering: 0.42 where the ratio is 0.97, and numerically zero once the ratio drops below about 0.7. The diamond is the optimal split of a single budget across all three vectors, which is budget-feasible and clears the line in four of the five scenarios.

Figure 9. *Expected* capacity of each vector at full effort, as a multiple of the manipulation threshold. A bar below the line means the mean falls short, not that the threshold is unreachable — the annotated *P* is the exact success probability at full effort. The diamond is the optimal split of a single budget across all three vectors, which is budget-feasible.

How to read it. Why a rational attacker declines a vector. Each bar is capacity as a multiple of the threshold *m*, so the threshold is the line at 1.0. It is an EXPECTED value, not a ceiling: a bar below the line means the mean falls short, and the annotated *P* gives the exact success probability at full effort — 0.42 at a ratio of 0.97, and numerically zero below about 0.7. The diamond is the optimal split of one budget across all three vectors, which is budget-feasible.



Protected coalition markers sit at zero in every one of these benchmark scenarios, under these parameter settings: the countermeasure configuration reduces the mixed attack to the same degree as it reduces each single vector. The wider randomised trials retain a non-zero residual, so this is a statement about these scenarios, not a general impossibility result.

Figure 10. Best coalition attack against best single-vector attack per scenario, with protected coalition results overlaid.

How to read it. The coalition result. The best mixed-vector attack against the best single vector, per scenario. Protected markers sit at zero across these benchmark scenarios under these parameter settings — a statement about those scenarios, not a general impossibility result, since the wider randomised trials retain a non-zero residual. Read the coalition analysis as an extension to the core baseline-versus-protected result, not as its centre.

Part II. Result Tables

The complete tabular output of the simulation, read directly from `results/csv/`. Column names are as emitted by the pipeline.

> **Which probability each table reports.** Sybil and ballot stuffing pass fraudulent votes through *two* independent gates, so they have two success probabilities. $P(\text{accepted} \geq m)$ counts votes that clear the *acceptance* gate only — an intermediate quantity that ignores detection and is therefore always larger. $P(\text{surviving} \geq m)$ counts votes that clear *both* gates, and is the headline measure used in Tables 7–9, Figure 4 and every stated conclusion. Bribery has one effective gate (detection is folded into $1-q$), so its two measures coincide. Tables 3 and 4 report both, labelled by a `measure` column; Tables 5 and 7–9 report the headline measure only, so the numbers agree across them.

Table 1. Parameter ranges used in the randomised trials

parameter	low	high
N	200.0000	5000.0000
m_fraction	0.0200	0.2500
B	1000.0000	200000.0000
R	10.0000	500.0000
P	10.0000	2000.0000
q_baseline	0.0200	0.2500
q_protected_boost	0.1500	0.6000
S	10.0000	1000.0000
qs_baseline	0.4000	0.9500
qs_protected_reduction	0.3000	0.8000
pd_baseline	0.0200	0.3000
pd_protected_boost	0.2000	0.7000
A	10.0000	1000.0000
qb_baseline	0.3000	0.9000
qb_protected_reduction	0.3000	0.8000
pb_baseline	0.0200	0.3000
pb_protected_boost	0.2000	0.7000

Table 2. Bribery attack — success probability (single gate: detection is folded into $1-q$)

attack	measure	system	mean	median	std	min	max
bribery	single_gate	baseline	0.3501	0.0000	0.4743	0.0000	1.0000
bribery	single_gate	protected	0.0635	0.0000	0.2421	0.0000	1.0000

Table 3. Sybil attack — both measures: surviving (headline) and accepted-only

attack	measure	system	mean	median	std	min	max
sybil	surviving_ge_m	baseline	0.4804	0.0593	0.4923	0.0000	1.0000
sybil	surviving_ge_m	protected	0.1356	0.0000	0.3292	0.0000	1.0000
sybil	accepted_ge_m	baseline	0.5366	0.9993	0.4931	0.0000	1.0000
sybil	accepted_ge_m	protected	0.2991	0.0000	0.4486	0.0000	1.0000

Table 4. Ballot stuffing — both measures: surviving (headline) and accepted-only

attack	measure	system	mean	median	std	min	max
ballot_stuffing	surviving_ge_m	baseline	0.4416	0.0000	0.4871	0.0000	1.0000
ballot_stuffing	surviving_ge_m	protected	0.1162	0.0000	0.3098	0.0000	1.0000
ballot_stuffing	accepted_ge_m	baseline	0.4979	0.4834	0.4926	0.0000	1.0000
ballot_stuffing	accepted_ge_m	protected	0.2535	0.0000	0.4239	0.0000	1.0000

Table 5. Baseline versus protected — headline measure only (agrees with Tables 7–9)

attack	measure	system	mean	median	std	min	max
bribery	single_gate	baseline	0.3501	0.0000	0.4743	0.0000	1.0000
bribery	single_gate	protected	0.0635	0.0000	0.2421	0.0000	1.0000
sybil	surviving_ge_m	baseline	0.4804	0.0593	0.4923	0.0000	1.0000
sybil	surviving_ge_m	protected	0.1356	0.0000	0.3292	0.0000	1.0000
ballot_stuffing	surviving_ge_m	baseline	0.4416	0.0000	0.4871	0.0000	1.0000
ballot_stuffing	surviving_ge_m	protected	0.1162	0.0000	0.3098	0.0000	1.0000

Table 6. Chernoff bound validation statistics

n valid	mean conservativeness	min conservativeness	max conservativeness	pct bound holds	attack
1088	0.0423	0.0000	0.8896	100.0000	bribery
3471	0.0260	0.0000	0.7550	100.0000	sybil
3604	0.0216	0.0000	0.7360	100.0000	ballot_stuffing

Table 7. Statistical analysis of the paired design

mean baseline	mean protected	absolute reduction	relative reduction	mean paired reduction	paired reduction std	bootstrap boot mean	bootstrap ci low	bootstrap ci high	attack type
0.3501	0.0635	0.2866	0.8186	0.2866	0.4483	0.2864	0.2689	0.3041	bribery
0.4804	0.1356	0.3448	0.7178	0.3448	0.4580	0.3445	0.3268	0.3622	sybil
0.4416	0.1162	0.3255	0.7370	0.3255	0.4510	0.3256	0.3084	0.3432	ballot_stuffing

Table 8. Countermeasure effectiveness (absolute and relative reduction)

attack type	mean baseline	mean protected	absolute reduction	relative reduction
bribery	0.3501	0.0635	0.2866	0.8186
sybil	0.4804	0.1356	0.3448	0.7178
ballot_stuffing	0.4416	0.1162	0.3255	0.7370

Table 9. Overall attack comparison across the three vectors

attack	mean baseline success	mean protected success	relative reduction pct	mc mean abs error	chernoff pct holds
bribery	0.3501	0.0635	81.8589	0.0001	100.0000
sybil	0.4804	0.1356	71.7790	0.0004	100.0000
ballot_stuffing	0.4416	0.1162	73.6953	0.0004	100.0000

Table 10. Sample scenarios with the manipulation threshold m

trial id	N	m	B	R	P	q	system type	exact probability	attack success exact
1	2266	275	122751.5049	257.4784	1479.1157	0.0842	baseline	1.0000	True
1	2266	275	122751.5049	257.4784	1479.1157	0.2615	protected	0.0000	False
2	4853	600	122396.0345	457.0915	1175.7973	0.1742	baseline	0.0000	False
2	4853	600	122396.0345	457.0915	1175.7973	0.6655	protected	0.0000	False
3	2832	325	48399.9988	339.8336	617.3681	0.1135	baseline	0.0000	False
3	2832	325	48399.9988	339.8336	617.3681	0.3816	protected	0.0000	False
4	2185	170	54426.8507	80.3503	1612.2060	0.0301	baseline	1.0000	True
4	2185	170	54426.8507	80.3503	1612.2060	0.2908	protected	0.0000	False
5	1339	28	32524.0897	268.5447	987.8435	0.1716	baseline	1.0000	True
5	1339	28	32524.0897	268.5447	987.8435	0.4312	protected	0.0000	False
6	4602	366	93406.3314	261.9435	1659.2226	0.2452	baseline	0.0000	False
6	4602	366	93406.3314	261.9435	1659.2226	0.3740	protected	0.0000	False
7	1215	176	53865.9070	342.5107	1767.2102	0.1220	baseline	0.0000	False
7	1215	176	53865.9070	342.5107	1767.2102	0.5104	protected	0.0000	False
8	301	42	16428.4483	497.2649	1961.4240	0.0719	baseline	0.0000	False
8	301	42	16428.4483	497.2649	1961.4240	0.2514	protected	0.0000	False
9	945	197	171607.6095	200.9098	417.8600	0.0833	baseline	1.0000	True
9	945	197	171607.6095	200.9098	417.8600	0.3743	protected	0.0000	False
10	3174	308	18530.7135	403.8856	1403.6775	0.2330	baseline	0.0000	False
10	3174	308	18530.7135	403.8856	1403.6775	0.3850	protected	0.0000	False
11	3779	710	98080.1864	64.3038	1942.2510	0.0745	baseline	0.0000	False
11	3779	710	98080.1864	64.3038	1942.2510	0.2938	protected	0.0000	False
12	4311	956	37465.0252	82.2381	131.7993	0.1598	baseline	0.0000	False
12	4311	956	37465.0252	82.2381	131.7993	0.5154	protected	0.0000	False
13	1855	287	181786.9165	89.5474	862.7410	0.1241	baseline	0.0000	False
13	1855	287	181786.9165	89.5474	862.7410	0.3897	protected	0.0000	False
14	3885	393	25016.9646	154.7345	476.1860	0.2479	baseline	0.0000	False
14	3885	393	25016.9646	154.7345	476.1860	0.4073	protected	0.0000	False
15	4092	973	55194.9407	206.1441	1075.9624	0.1898	baseline	0.0000	False
15	4092	973	55194.9407	206.1441	1075.9624	0.3552	protected	0.0000	False

Table 11. Critical thresholds T1–T7 per benchmark scenario

scenario id	name	system type	N	m	B	R	P	q	R star	q voter indifference	voter accepts	T feasible	T star	T min for even odds	B star	q defender critical	mu	m reachable	m certified	exact p flip	utility star
S1	Small university election (100 voters)	baseline	100	35	3000	80	600	0.0800	52.1739	0.1176	True	37	37	38.0000	2880.0000	0.1176	34.0400	34.0000	NaN	0.4231	21471.9356
S1	Small university election (100 voters)	protected	100	35	3000	80	600	0.5500	733.3333	0.1176	False	37	0	77.0000	NaN	0.1176	16.6500	17.0000	34.0000	0.0000	0.0000
S2	Medium organisation election (10,000 voters)	baseline	10000	1943	250000	120	900	0.0700	67.7419	0.1176	True	2083	2083	2089.0000	249480.0002	0.0722	1937.1900	1937.0000	NaN	0.3277	255503.1621
S2	Medium organisation election (10,000 voters)	protected	10000	1943	250000	120	900	0.5500	1100.0000	0.1176	False	2083	0	4317.0000	NaN	0.0722	937.3500	937.0000	1054.0000	0.0000	0.0000
S3	Large national-scale election (100,000 voters)	baseline	100000	19015	3000000	150	1200	0.0500	63.1579	0.1111	True	20000	20000	20016.0000	2998649.9998	0.0504	19000.0000	19000.0000	19515.0000	0.3204	1907536.0062
S3	Large national-scale election (100,000 voters)	protected	100000	19015	3000000	150	1200	0.5500	1466.6667	0.1111	False	20000	0	42255.0000	NaN	0.0504	9000.0000	9000.0000	NaN	0.0000	0.0000
S4	High-budget attack (maximum attacker capability)	baseline	5000	4758	5000000	100	500	0.0500	26.3158	0.1667	True	5000	5000	5008.0000	499400.0004	0.0514	4750.0000	4750.0000	NaN	0.3159	679480.1805
S4	High-budget attack (maximum attacker capability)	protected	5000	4758	5000000	100	500	0.5500	611.1111	0.1667	False	5000	0	10573.0000	NaN	0.0514	2250.0000	2250.0000	NaN	0.0000	0.0000
S5	Low-budget attack (expected to fail)	baseline	5000	550	8000	100	500	0.0500	26.3158	0.1667	True	80	0	579.0000	57100.0004	0.0000	76.0000	76.0000	NaN	0.0000	0.0000
S5	Low-budget attack (expected to fail)	protected	5000	550	8000	100	500	0.5500	611.1111	0.1667	False	80	0	1222.0000	NaN	0.0000	36.0000	36.0000	61.0000	0.0000	0.0000

Table 12. Optimal attacker effort per vector (G1)

scenario id	name	system type	vector	unit cost	p survive	p detect	m	n feasible	n star	p success at star	cost at star	utility star	viable	gate open
S1	Small university election (100 voters)	baseline	bribery	80.0000	0.9200	0.0800	35	37	37	0.4231	2960.0000	21471.9356	True	True
S1	Small university election (100 voters)	baseline	sybil	32.0000	0.7200	0.1000	35	20	0	0.0000	0.0000	0.0000	False	True
S1	Small university election (100 voters)	baseline	ballot_stuffing	12.0000	0.6300	0.1000	35	30	0	0.0000	0.0000	0.0000	False	True
S1	Small university election (100 voters)	protected	bribery	80.0000	0.4500	0.5500	35	37	0	0.0000	0.0000	0.0000	False	False
S1	Small university election (100 voters)	protected	sybil	32.0000	0.1125	0.5500	35	20	0	0.0000	0.0000	0.0000	False	True
S1	Small university election (100 voters)	protected	ballot_stuffing	12.0000	0.0900	0.5500	35	30	0	0.0000	0.0000	0.0000	False	True
S2	Medium organisation election (10,000 voters)	baseline	bribery	120.0000	0.9300	0.0700	1943	2083	2083	0.3277	249960.0000	255503.1621	True	True
S2	Medium organisation election (10,000 voters)	baseline	sybil	48.0000	0.7200	0.1000	1943	2000	0	0.0000	0.0000	0.0000	False	True
S2	Medium organisation election (10,000 voters)	baseline	ballot_stuffing	18.0000	0.6300	0.1000	1943	3000	0	0.0000	0.0000	0.0000	False	True
S2	Medium organisation election (10,000 voters)	protected	bribery	120.0000	0.4500	0.5500	1943	2083	0	0.0000	0.0000	0.0000	False	False
S2	Medium organisation election (10,000 voters)	protected	sybil	48.0000	0.1125	0.5500	1943	2000	0	0.0000	0.0000	0.0000	False	True
S2	Medium organisation election (10,000 voters)	protected	ballot_stuffing	18.0000	0.0900	0.5500	1943	3000	0	0.0000	0.0000	0.0000	False	True
S3	Large national-scale election (100,000 voters)	baseline	bribery	150.0000	0.9500	0.0500	19015	20000	20000	0.3204	3000000.0000	1907536.0062	True	True
S3	Large national-scale election (100,000 voters)	baseline	sybil	60.0000	0.7200	0.1000	19015	20000	0	0.0000	0.0000	0.0000	False	True
S3	Large national-scale election (100,000 voters)	baseline	ballot_stuffing	22.5000	0.6300	0.1000	19015	30000	0	0.0000	0.0000	0.0000	False	True
S3	Large national-scale election (100,000 voters)	protected	bribery	150.0000	0.4500	0.5500	19015	20000	0	0.0000	0.0000	0.0000	False	False
S3	Large national-scale election (100,000 voters)	protected	sybil	60.0000	0.1125	0.5500	19015	20000	0	0.0000	0.0000	0.0000	False	True
S3	Large national-scale election (100,000 voters)	protected	ballot_stuffing	22.5000	0.0900	0.5500	19015	30000	0	0.0000	0.0000	0.0000	False	True
S4	High-budget attack (maximum attacker capability)	baseline	bribery	100.0000	0.9500	0.0500	4758	5000	5000	0.3159	500000.0000	679480.1805	True	True
S4	High-budget attack (maximum attacker capability)	baseline	sybil	40.0000	0.7200	0.1000	4758	1000	0	0.0000	0.0000	0.0000	False	True
S4	High-budget attack (maximum attacker capability)	baseline	ballot_stuffing	15.0000	0.6300	0.1000	4758	1500	0	0.0000	0.0000	0.0000	False	True
S4	High-budget attack (maximum attacker capability)	protected	bribery	100.0000	0.4500	0.5500	4758	5000	0	0.0000	0.0000	0.0000	False	False
S4	High-budget attack (maximum attacker capability)	protected	sybil	40.0000	0.1125	0.5500	4758	1000	0	0.0000	0.0000	0.0000	False	True
S4	High-budget attack (maximum attacker capability)	protected	ballot_stuffing	15.0000	0.0900	0.5500	4758	1500	0	0.0000	0.0000	0.0000	False	True
S5	Low-budget attack (expected to fail)	baseline	bribery	100.0000	0.9500	0.0500	550	80	0	0.0000	0.0000	0.0000	False	True
S5	Low-budget attack (expected to fail)	baseline	sybil	40.0000	0.7200	0.1000	550	200	0	0.0000	0.0000	0.0000	False	True
S5	Low-budget attack (expected to fail)	baseline	ballot_stuffing	15.0000	0.6300	0.1000	550	533	0	0.0000	0.0000	0.0000	False	True
S5	Low-budget attack (expected to fail)	protected	bribery	100.0000	0.4500	0.5500	550	80	0	0.0000	0.0000	0.0000	False	False
S5	Low-budget attack (expected to fail)	protected	sybil	40.0000	0.1125	0.5500	550	200	0	0.0000	0.0000	0.0000	False	True
S5	Low-budget attack (expected to fail)	protected	ballot_stuffing	15.0000	0.0900	0.5500	550	533	0	0.0000	0.0000	0.0000	False	True

Table 13. Coalition attack versus best single vector (G2)

scenario id	name	system type	m	B	share bribery	share sybil	share ballot stuffing	n bribery	n sybil	n ballot stuffing	mu total	coalition exact p	coalition chernoff	coalition p noticed	coalition utility	best pure vector	best pure exact p	best pure utility	coalition gain p	coalition gain utility
S1	Small university election (100 voters)	baseline	35	3000	0.3500	0.2500	0.4000	13	20	30	45.2600	0.9987	NaN	0.9983	56881.5449	bribery	0.4231	21471.9356	0.5756	35409.6093
S1	Small university election (100 voters)	protected	35	3000	1.0000	0.0000	0.0000	0	0	0	0.0000	0.0000	NaN	0.0000	0.0000	bribery	0.0000	0.0000	0.0000	0.0000
S2	Medium organisation election (10,000 voters)	baseline	1943	250000	0.0000	0.0500	0.9500	0	260	3000	2077.2000	1.0000	NaN	1.0000	1783518.9291	bribery	0.3277	255503.1621	0.6723	1528015.7670
S2	Medium organisation election (10,000 voters)	protected	1943	250000	1.0000	0.0000	0.0000	0	0	0	0.0000	0.0000	NaN	0.0000	0.0000	bribery	0.0000	0.0000	0.0000	0.0000
S3	Large national-scale election (100,000 voters)	baseline	19015	3000000	0.0000	0.0500	0.9500	0	2500	30000	20700.0000	1.0000	NaN	1.0000	17675000.0000	bribery	0.3204	1907536.0062	0.6796	15767463.9938
S3	Large national-scale election (100,000 voters)	protected	19015	3000000	1.0000	0.0000	0.0000	0	0	0	0.0000	0.0000	NaN	0.0000	0.0000	bribery	0.0000	0.0000	0.0000	0.0000
S4	High-budget attack (maximum attacker capability)	baseline	4758	5000000	0.1000	0.0000	0.9000	5000	0	1500	5695.0000	1.0000	NaN	1.0000	4077500.0000	bribery	0.3159	679480.1805	0.6841	3398019.8195
S4	High-budget attack (maximum attacker capability)	protected	4758	5000000	1.0000	0.0000	0.0000	0	0	0	0.0000	0.0000	NaN	0.0000	0.0000	bribery	0.0000	0.0000	0.0000	0.0000
S5	Low-budget attack (expected to fail)	baseline	550	8000	1.0000	0.0000	0.0000	80	0	0	76.0000	0.0000	0.0000	0.9835	-106348.4626	bribery	0.0000	-106348.4626	0.0000	0.0000
S5	Low-budget attack (expected to fail)	protected	550	8000	1.0000	0.0000	0.0000	0	0	0	0.0000	0.0000	NaN	0.0000	0.0000	bribery	0.0000	0.0000	0.0000	0.0000

Table 14. Sensitivity of the coalition finding to cost assumptions

scenario id	identity cost ratio	ballot cost ratio	sybil ceiling fraction	ballot ceiling fraction	coalition exact p	best pure exact p	coalition utility	best pure utility	coalition at least as good	coalition strictly better
S1	0.2000	0.0500	0.1000	0.1500	0.9995	0.4231	56356.0926	21471.9356	True	True
S2	0.2000	0.0500	0.1000	0.1500	1.0000	0.3277	1767073.3386	255503.1621	True	True
S3	0.2000	0.0500	0.1000	0.1500	1.0000	0.3204	17487500.0000	1907536.0062	True	True
S1	0.2000	0.0500	0.1000	0.3000	0.9997	0.4231	56947.0209	21471.9356	True	True
S2	0.2000	0.0500	0.1000	0.3000	1.0000	0.3277	1808000.0000	255503.1621	True	True
S3	0.2000	0.0500	0.1000	0.3000	1.0000	0.3204	17975000.0000	1907536.0062	True	True
S1	0.2000	0.0500	0.2000	0.1500	0.9987	0.4231	56787.6301	21471.9356	True	True
S2	0.2000	0.0500	0.2000	0.1500	1.0000	0.3277	1793000.0000	255503.1621	True	True
S3	0.2000	0.0500	0.2000	0.1500	1.0000	0.3204	17787500.0000	1907536.0062	True	True
S1	0.2000	0.0500	0.2000	0.3000	0.9997	0.4231	57346.1828	21471.9356	True	True
S2	0.2000	0.0500	0.2000	0.3000	1.0000	0.3277	1807039.8376	255503.1621	True	True
S3	0.2000	0.0500	0.2000	0.3000	1.0000	0.3204	17975000.0000	1907536.0062	True	True
S1	0.2000	0.1500	0.1000	0.1500	0.9995	0.4231	56236.0926	21471.9356	True	True
S2	0.2000	0.1500	0.1000	0.1500	1.0000	0.3277	1749073.3386	255503.1621	True	True
S3	0.2000	0.1500	0.1000	0.1500	1.0000	0.3204	17300006.0941	1907536.0062	True	True
S1	0.2000	0.1500	0.1000	0.3000	0.9997	0.4231	56707.0209	21471.9356	True	True
S2	0.2000	0.1500	0.1000	0.3000	1.0000	0.3277	1776014.0000	255503.1621	True	True
S3	0.2000	0.1500	0.1000	0.3000	1.0000	0.3204	17600015.0000	1907536.0062	True	True
S1	0.2000	0.1500	0.2000	0.1500	0.9987	0.4231	56667.6301	21471.9356	True	True
S2	0.2000	0.1500	0.2000	0.1500	1.0000	0.3277	1777016.0000	255503.1621	True	True
S3	0.2000	0.1500	0.2000	0.1500	1.0000	0.3204	17600007.5000	1907536.0062	True	True
S1	0.2000	0.1500	0.2000	0.3000	0.9997	0.4231	57106.1828	21471.9356	True	True
S2	0.2000	0.1500	0.2000	0.3000	1.0000	0.3277	1777016.0000	255503.1621	True	True
S3	0.2000	0.1500	0.2000	0.3000	1.0000	0.3204	17600007.5000	1907536.0062	True	True
S1	0.2000	0.4000	0.1000	0.1500	0.9881	0.4231	55446.3963	21471.9356	True	True
S2	0.2000	0.4000	0.1000	0.1500	1.0000	0.3277	1704073.3386	255503.1621	True	True
S3	0.2000	0.4000	0.1000	0.1500	1.0000	0.3204	16700000.0000	1907536.0062	True	True
S1	0.2000	0.4000	0.1000	0.3000	0.9993	0.4231	56142.9381	21471.9356	True	True
S2	0.2000	0.4000	0.1000	0.3000	0.9997	0.3277	1725355.1119	255503.1621	True	True
S3	0.2000	0.4000	0.1000	0.3000	1.0000	0.3204	17000000.0000	1907536.0062	True	True
S1	0.2000	0.4000	0.2000	0.1500	0.9987	0.4231	56367.6301	21471.9356	True	True
S2	0.2000	0.4000	0.2000	0.1500	1.0000	0.3277	1752031.9974	255503.1621	True	True
S3	0.2000	0.4000	0.2000	0.1500	1.0000	0.3204	17300000.0000	1907536.0062	True	True
S1	0.2000	0.4000	0.2000	0.3000	0.9993	0.4231	56543.9245	21471.9356	True	True
S2	0.2000	0.4000	0.2000	0.3000	1.0000	0.3277	1752031.9974	255503.1621	True	True
S3	0.2000	0.4000	0.2000	0.3000	1.0000	0.3204	17300000.0000	1907536.0062	True	True
S1	0.4000	0.0500	0.1000	0.1500	0.9990	0.4231	56200.3147	21471.9356	True	True
S2	0.4000	0.0500	0.1000	0.1500	1.0000	0.3277	1743073.3386	255503.1621	True	True
S3	0.4000	0.0500	0.1000	0.1500	1.0000	0.3204	17187500.0000	1907536.0062	True	True
S1	0.4000	0.0500	0.1000	0.3000	0.9995	0.4231	56805.7090	21471.9356	True	True
S2	0.4000	0.0500	0.1000	0.3000	1.0000	0.3277	1807040.0000	255503.1621	True	True
S3	0.4000	0.0500	0.1000	0.3000	1.0000	0.3204	17975000.0000	1907536.0062	True	True
S1	0.4000	0.0500	0.2000	0.1500	0.9987	0.4231	56467.6301	21471.9356	True	True
S2	0.4000	0.0500	0.2000	0.1500	1.0000	0.3277	1766022.9615	255503.1621	True	True
S3	0.4000	0.0500	0.2000	0.1500	1.0000	0.3204	17487500.0000	1907536.0062	True	True
S1	0.4000	0.0500	0.2000	0.3000	0.9992	0.4231	57055.3374	21471.9356	True	True
S2	0.4000	0.0500	0.2000	0.3000	1.0000	0.3277	1807040.0000	255503.1621	True	True

Continued on next page

scenario id	identity cost ratio	ballot cost ratio	sybil ceiling fraction	ballot ceiling fraction	coalition exact p	best pure exact p	coalition utility	best pure utility	coalition at least as good	coalition strictly better
S3	0.4000	0.0500	0.2000	0.3000	1.0000	0.3204	17975000.0000	1907536.0062	True	True
S1	0.4000	0.1500	0.1000	0.1500	0.9990	0.4231	56080.3147	21471.9356	True	True
S2	0.4000	0.1500	0.1000	0.1500	1.0000	0.3277	1725073.3386	255503.1621	True	True
S3	0.4000	0.1500	0.1000	0.1500	1.0000	0.3204	17000006.0941	1907536.0062	True	True
S1	0.4000	0.1500	0.1000	0.3000	0.9995	0.4231	56565.7090	21471.9356	True	True
S2	0.4000	0.1500	0.1000	0.3000	1.0000	0.3277	1771040.0000	255503.1621	True	True
S3	0.4000	0.1500	0.1000	0.3000	1.0000	0.3204	17525000.0000	1907536.0062	True	True
S1	0.4000	0.1500	0.2000	0.1500	0.9987	0.4231	56347.6301	21471.9356	True	True
S2	0.4000	0.1500	0.2000	0.1500	1.0000	0.3277	1748022.9615	255503.1621	True	True
S3	0.4000	0.1500	0.2000	0.1500	1.0000	0.3204	17262500.0000	1907536.0062	True	True
S1	0.4000	0.1500	0.2000	0.3000	0.9992	0.4231	56815.3374	21471.9356	True	True
S2	0.4000	0.1500	0.2000	0.3000	1.0000	0.3277	1771040.0000	255503.1621	True	True
S3	0.4000	0.1500	0.2000	0.3000	1.0000	0.3204	17525000.0000	1907536.0062	True	True
S1	0.4000	0.4000	0.1000	0.1500	0.9789	0.4231	54771.5502	21471.9356	True	True
S2	0.4000	0.4000	0.1000	0.1500	1.0000	0.3277	1680073.3386	255503.1621	True	True
S3	0.4000	0.4000	0.1000	0.1500	1.0000	0.3204	16400000.0000	1907536.0062	True	True
S1	0.4000	0.4000	0.1000	0.3000	0.9987	0.4231	55978.5555	21471.9356	True	True
S2	0.4000	0.4000	0.1000	0.3000	0.9997	0.3277	1701355.1119	255503.1621	True	True
S3	0.4000	0.4000	0.1000	0.3000	1.0000	0.3204	16700000.0000	1907536.0062	True	True
S1	0.4000	0.4000	0.2000	0.1500	0.9958	0.4231	55939.2111	21471.9356	True	True
S2	0.4000	0.4000	0.2000	0.1500	1.0000	0.3277	1704031.9974	255503.1621	True	True
S3	0.4000	0.4000	0.2000	0.1500	1.0000	0.3204	16700000.0000	1907536.0062	True	True
S1	0.4000	0.4000	0.2000	0.3000	0.9993	0.4231	56223.9245	21471.9356	True	True
S2	0.4000	0.4000	0.2000	0.3000	1.0000	0.3277	1704031.9974	255503.1621	True	True
S3	0.4000	0.4000	0.2000	0.3000	1.0000	0.3204	16700000.0000	1907536.0062	True	True
S1	0.8000	0.0500	0.1000	0.1500	0.9816	0.4231	55210.7680	21471.9356	True	True
S2	0.8000	0.0500	0.1000	0.1500	1.0000	0.3277	1695073.3386	255503.1621	True	True
S3	0.8000	0.0500	0.1000	0.1500	1.0000	0.3204	16587500.0000	1907536.0062	True	True
S1	0.8000	0.0500	0.1000	0.3000	0.9964	0.4231	56587.8295	21471.9356	True	True
S2	0.8000	0.0500	0.1000	0.3000	1.0000	0.3277	1807039.8376	255503.1621	True	True
S3	0.8000	0.0500	0.1000	0.3000	1.0000	0.3204	17975000.0000	1907536.0062	True	True
S1	0.8000	0.0500	0.2000	0.1500	0.9816	0.4231	55210.7680	21471.9356	True	True
S2	0.8000	0.0500	0.2000	0.1500	1.0000	0.3277	1691143.9853	255503.1621	True	True
S3	0.8000	0.0500	0.2000	0.1500	1.0000	0.3204	16587500.0000	1907536.0062	True	True
S1	0.8000	0.0500	0.2000	0.3000	0.9964	0.4231	56587.8295	21471.9356	True	True
S2	0.8000	0.0500	0.2000	0.3000	1.0000	0.3277	1807039.8376	255503.1621	True	True
S3	0.8000	0.0500	0.2000	0.3000	1.0000	0.3204	17975000.0000	1907536.0062	True	True
S1	0.8000	0.1500	0.1000	0.1500	0.9816	0.4231	55090.7680	21471.9356	True	True
S2	0.8000	0.1500	0.1000	0.1500	1.0000	0.3277	1677073.3386	255503.1621	True	True
S3	0.8000	0.1500	0.1000	0.1500	1.0000	0.3204	16400006.0941	1907536.0062	True	True
S1	0.8000	0.1500	0.1000	0.3000	0.9964	0.4231	56347.8295	21471.9356	True	True
S2	0.8000	0.1500	0.1000	0.3000	1.0000	0.3277	1771039.8376	255503.1621	True	True
S3	0.8000	0.1500	0.1000	0.3000	1.0000	0.3204	17525000.0000	1907536.0062	True	True
S1	0.8000	0.1500	0.2000	0.1500	0.9816	0.4231	55090.7680	21471.9356	True	True
S2	0.8000	0.1500	0.2000	0.1500	1.0000	0.3277	1673143.9853	255503.1621	True	True
S3	0.8000	0.1500	0.2000	0.1500	1.0000	0.3204	16362500.0000	1907536.0062	True	True
S1	0.8000	0.1500	0.2000	0.3000	0.9964	0.4231	56347.8295	21471.9356	True	True
S2	0.8000	0.1500	0.2000	0.3000	1.0000	0.3277	1771039.8376	255503.1621	True	True
S3	0.8000	0.1500	0.2000	0.3000	1.0000	0.3204	17525000.0000	1907536.0062	True	True
S1	0.8000	0.4000	0.1000	0.1500	0.8571	0.4231	47560.7561	21471.9356	True	True
S2	0.8000	0.4000	0.1000	0.1500	1.0000	0.3277	1632073.3386	255503.1621	True	True

Continued on next page

scenario id	identity cost ratio	ballot cost ratio	sybil ceiling fraction	ballot ceiling fraction	coalition exact p	best pure exact p	coalition utility	best pure utility	coalition at least as good	coalition strictly better
S3	0.8000	0.4000	0.1000	0.1500	1.0000	0.3204	15800000.0000	1907536.0062	True	True
S1	0.8000	0.4000	0.1000	0.3000	0.9906	0.4231	55467.9365	21471.9356	True	True
S2	0.8000	0.4000	0.1000	0.3000	1.0000	0.3277	1681039.8376	255503.1621	True	True
S3	0.8000	0.4000	0.1000	0.3000	1.0000	0.3204	16400000.0000	1907536.0062	True	True
S1	0.8000	0.4000	0.2000	0.1500	0.8571	0.4231	47560.7561	21471.9356	True	True
S2	0.8000	0.4000	0.2000	0.1500	1.0000	0.3277	1628143.9853	255503.1621	True	True
S3	0.8000	0.4000	0.2000	0.1500	1.0000	0.3204	15800000.0000	1907536.0062	True	True
S1	0.8000	0.4000	0.2000	0.3000	0.9906	0.4231	55467.9365	21471.9356	True	True
S2	0.8000	0.4000	0.2000	0.3000	1.0000	0.3277	1681039.8376	255503.1621	True	True
S3	0.8000	0.4000	0.2000	0.3000	1.0000	0.3204	16400000.0000	1907536.0062	True	True

Part III. Headline Numbers and Reproducibility

Scale of the experiment

- Paired trials: 2,500, with 2,000 Monte Carlo draws per trial
- Random seed: 42 — re-running reproduces every number bit-for-bit
- Automated tests: 119, all passing
- Figures: 10

Reproducibility protocol

All randomness derives from a single seeded `numpy.random.Generator`; no global random state and no unseeded standard-library random calls exist in the codebase. The threshold and coalition analyses are fully analytic and require no seed at all.

Every number in this document is injected at build time from `results/json/` and `results/csv/` as a LaTeX macro. A macro whose experiment has not been run causes the document to fail compilation rather than to print a stale value, which makes it structurally impossible for the text to disagree with the data it describes. `check_consistency.py` additionally verifies that claimed test and figure counts match reality.

How to regenerate this document

```
python3 main.py                # run the full simulation
python3 generate_figures.py    # write the 10 figures
python3 generate_tables.py     # write the 14 result tables
python3 build_compendium.py    # build this compendium
```